



CENG424 Embedded Computer Systems
Homework 4
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1. Introduction

This report presents the design and simulation of a PID (Proportional-Integral-Derivative) controller for a servo motor position control system. The objective is to drive the motor's angular position from 0 radians to the desired reference position of π radians (180°) while achieving satisfactory transient and steady-state performance.

The servo motor dynamics are governed by the following second-order differential equation:

$$J \cdot d^2\theta(t)/dt^2 + B \cdot d\theta(t)/dt = K_t \cdot u(t)$$

where $\theta(t)$ is the angular position (radians), $u(t)$ is the control input voltage, J is the moment of inertia, B is the damping coefficient, and K_t is the torque constant.

2. System Parameters

The simulation uses the following motor parameters defined in the hw4_parameters.m script:

Parameter	Value	Unit
Moment of Inertia (J)	0.005	kg·m ²
Damping Coefficient (B)	0.6	N·m·s/rad
Torque Constant (K _t)	0.08	N·m/A
Desired Position (θ_{desired})	$\pi \approx 3.1416$	rad (180°)
Initial Position (θ_0)	0	rad
K1 = K _t /J	16	-
K2 = B/J	120	-

3. Simulink Model

The servo motor system was modeled in MATLAB/Simulink using the given second-order differential equation:

$$J\ddot{\theta}(t) + B\dot{\theta}(t) = K_t u(t)$$

By rearranging the equation, the motor acceleration can be written as

$$\ddot{\theta}(t) = (K_t/J)u(t) - (B/J)\dot{\theta}(t)$$

Using the given parameters, the constants are obtained as $K1 = K_t/J = 16$ and $K2 = B/J = 120$. Based on this form, the plant was implemented in Simulink using Gain, Sum, and Integrator blocks. The first integrator generates the angular velocity, and the second integrator generates the angular position. The output position $\theta(t)$ is fed back and compared with the desired reference position $\theta_{\text{desired}} = \pi$ rad to produce the error signal. This error is then processed by the PID controller to generate the control input $u(t)$. The complete Simulink model is shown in Figure 1.

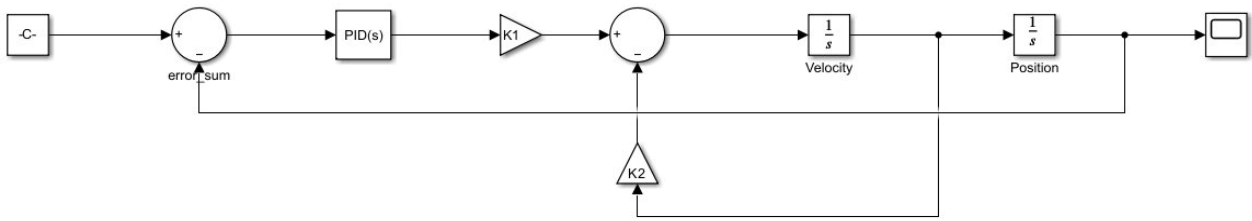


Figure 1: Simulink Block Diagram – Servo Motor PID Control System

4. PID Controller Design and Tuning

The PID Tuner was applied to the plant model (Type: PIDF, Form: Parallel, Domain: Time). The tuning result and the corresponding step response are shown below.

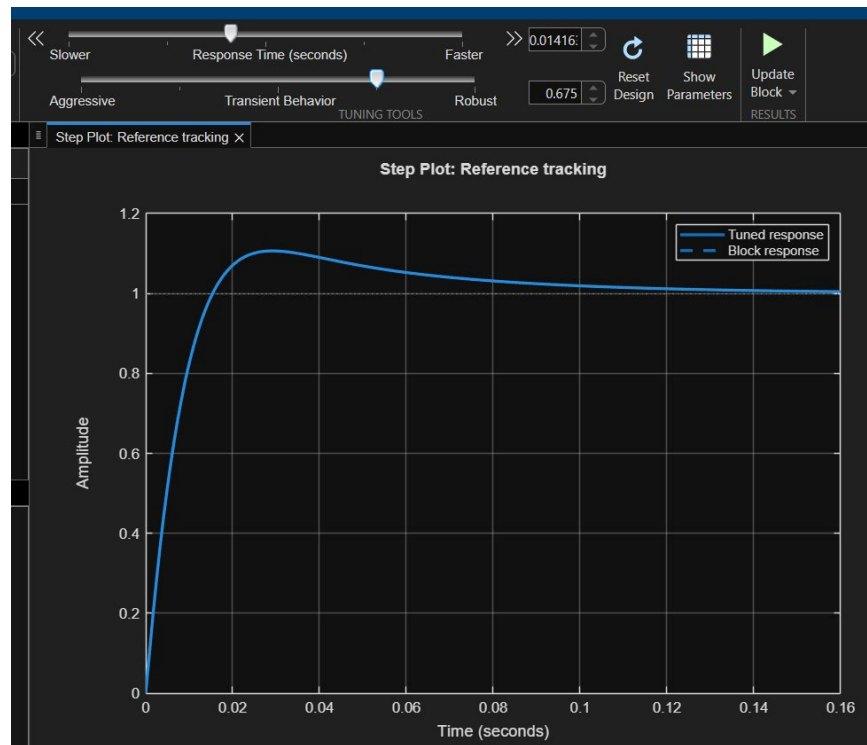


Figure 2: PID Tuner – Step Plot: Reference Tracking

Controller Parameters			
	Tuned	Block	
P	1310.853	1310.853	▲
I	23747.765	23747.765	
D	8.0215	8.0215	
N	16137.2607	16137.2607	
			▼
Performance and Robustness			
	Tuned	Block	
Rise time	0.011 seconds	0.011 seconds	▲
Settling time	0.0979 seconds	0.0979 seconds	
Overshoot	10.6 %	10.6 %	
Peak	1.11	1.11	
Gain margin	-Inf dB @ 0 rad/s	-Inf dB @ 0 rad/s	
Phase margin	76.5 deg @ 141 rad/s	76.5 deg @ 141 rad/s	
Closed-loop stability	Stable	Stable	▼

Figure 3: PID Tuner – Controller Parameters and Performance Metrics

The tuned controller parameters obtained from the PID Tuner are summarized in the table below.

Parameter	Value
Proportional Gain (P)	1310.853
Integral Gain (I)	23747.765
Derivative Gain (D)	8.0215
Filter Coefficient (N)	16137.2607

5. Simulation Results

The Simulink model was simulated to observe the position response of the servo motor under PID control. The output position was monitored using a Scope block. As shown in Figure 4, the motor position starts from 0 rad, rises rapidly toward the reference value of π rad, exhibits a small overshoot, and then settles to the desired position. The simulation results are consistent with the PID Tuner prediction and confirm that the designed controller provides both accurate tracking and fast transient response.

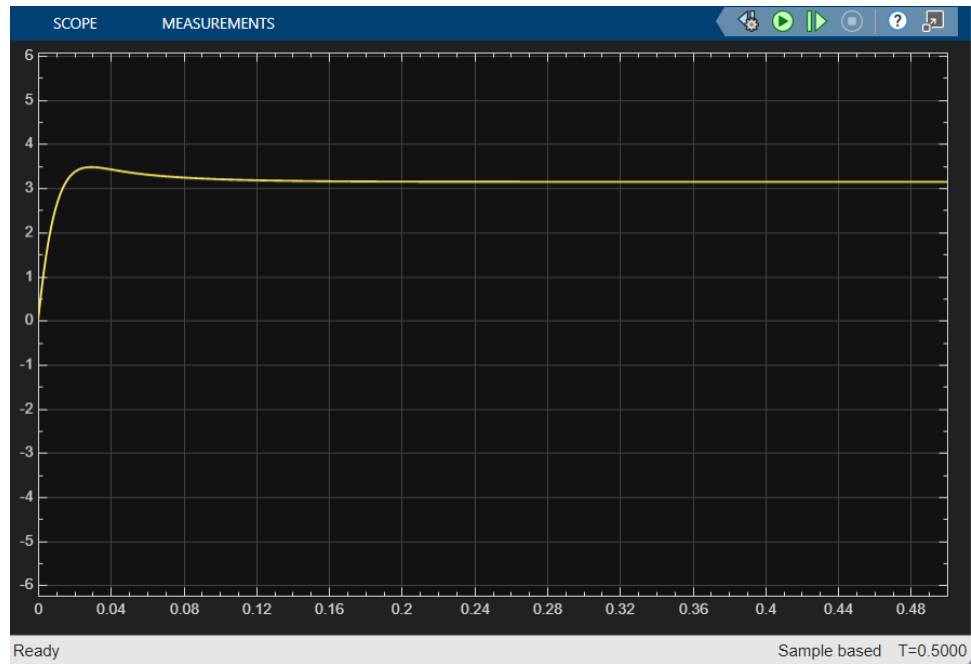


Figure 4: Simulink Scope Output – Angular Position $\theta(t)$ vs. Time

5.1 Performance Summary

Metric	Value
Rise Time	0.011 seconds
Settling Time	0.0979 seconds
Overshoot	10.6 %
Peak Value	$1.11 \times \pi \approx 3.49$ rad
Phase Margin	76.5° @ 141 rad/s
Closed-loop Stability	Stable

6. Performance Evaluation

6.1 Position Accuracy

The system achieves accurate steady-state tracking of the desired position. The output converges to approximately π rad, which indicates that the steady-state error is essentially zero. This is mainly due to the integral action of the controller, which eliminates residual tracking error over time and ensures precise final positioning.

6.2 Response Time

The designed controller provides a fast transient response. According to the PID Tuner results, the rise time is 0.011 s and the settling time is approximately 0.0979 s. This means that the motor reaches the target position and settles in less than 0.1 seconds, which indicates a highly responsive control system for the given plant parameters.

6.3 Overshoot and Stability

The closed-loop system exhibits an overshoot of 10.6%, meaning the motor briefly exceeds the target angle before settling. For many servo motor applications, this level of overshoot is acceptable; however, in precision positioning systems where mechanical stops or delicate loads are involved, it could be reduced by adjusting the Transient Behavior slider toward Robust in the PID Tuner. The phase margin of 76.5° at 141 rad/s is well above the commonly accepted threshold of 45° , indicating a robust and stable closed-loop system with good disturbance rejection.

The reported gain margin of $-\infty$ dB at 0 rad/s is related to the integrator in the loop and should not be interpreted alone as a sign of instability. The positive phase margin and the stable step response indicate that the closed-loop system is stable.

7. Conclusion

In this study, a PID-controlled servo motor position control system was successfully modeled and simulated in MATLAB/Simulink. The plant was derived from the given second-order motor equation and implemented using gain, summation, and integrator blocks. The PID controller was tuned using MATLAB PID Tuner, and the resulting controller provided fast and accurate position tracking.

The simulation results show that the motor position reaches the desired reference of π rad with nearly zero steady-state error, a rise time of 0.011 s, a settling time of approximately 0.098 s, and an overshoot of 10.6%. Both the transient response and the stability margins indicate that the designed controller performs effectively for the given servo motor parameters. Overall, the control objective was achieved successfully.